Magneto-static boundary conditions

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Scalar and Vector magnetic potentials

- Scalar magnetic potential – $\mathbf{H} = -\nabla V_m$ Similar to $\mathbf{E} = -\nabla V$
- Vector magnetic potential – $\mathbf{B} = \nabla \times \mathbf{A}$
- Reason: Gauss's law for magnetic field is $\nabla \cdot \mathbf{B} = 0$ ---(1)
- WKT vector identity The divergence of the curl of *any* vector field **A** is always zero: $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ --(2)
- Comparing (1) and (2) $\mathbf{B} = \nabla \times \mathbf{A}$

Note:

- A problem can be asked – given A find B
- Solution: Find Curl of A to obtain B

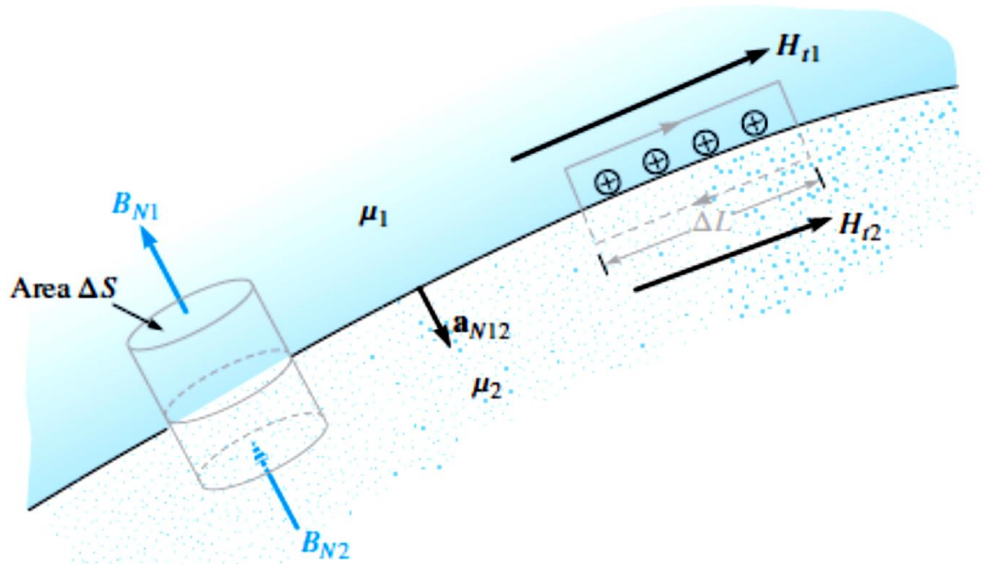
Curl Formula in Cylindrical

$$\nabla \times \vec{A} = \frac{1}{\rho}\begin{vmatrix} \hat{a}_\rho & \rho\hat{a}_\phi & \hat{a}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$

Curl Formula in Spherical

$$\nabla \times \vec{A} = \frac{1}{r^2 \sin\theta}\begin{vmatrix} \hat{a}_r & r\hat{a}_\theta & r\sin\theta\hat{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & rA_\theta & r\sin\theta A_\phi \end{vmatrix}$$

MAGNETIC BOUNDARY CONDITIONS



Electrostatics

$$E_{\tan 1} = E_{\tan 2}$$

$$D_{N1} - D_{N2} = \rho_S$$

$$B_{N2} = B_{N1}$$

$$(\mathbf{B}_2 - \mathbf{B}_1) \cdot \mathbf{a}_{N12} = 0$$

$$H_{t1} - H_{t2} = K$$

$$(\mathbf{H}_1 - \mathbf{H}_2) \times \mathbf{a}_{N12} = \mathbf{K}$$

A Gaussian surface and a closed path are constructed at the boundary between media 1 and 2, having permeabilities of μ_1 and μ_2, respectively.

$$\oint_S B \cdot d\mathbf{S} = 0$$

$$B_{N1}\Delta S - B_{N2}\Delta S = 0$$

$$B_{N2} = B_{N1} \quad \text{and} \quad H_{N2} = \frac{\mu_1}{\mu_2} H_{N1}$$

$$\oint \mathbf{H} \cdot d\mathbf{L} = I$$

$$H_{t1}\Delta L - H_{t2}\Delta L = K\Delta L$$

$$H_{t1} - H_{t2} = K \quad \text{and} \quad \frac{B_{t1}}{\mu_1} - \frac{B_{t2}}{\mu_2} = K$$

Practice this example from the textbook

EXAMPLE 8.6

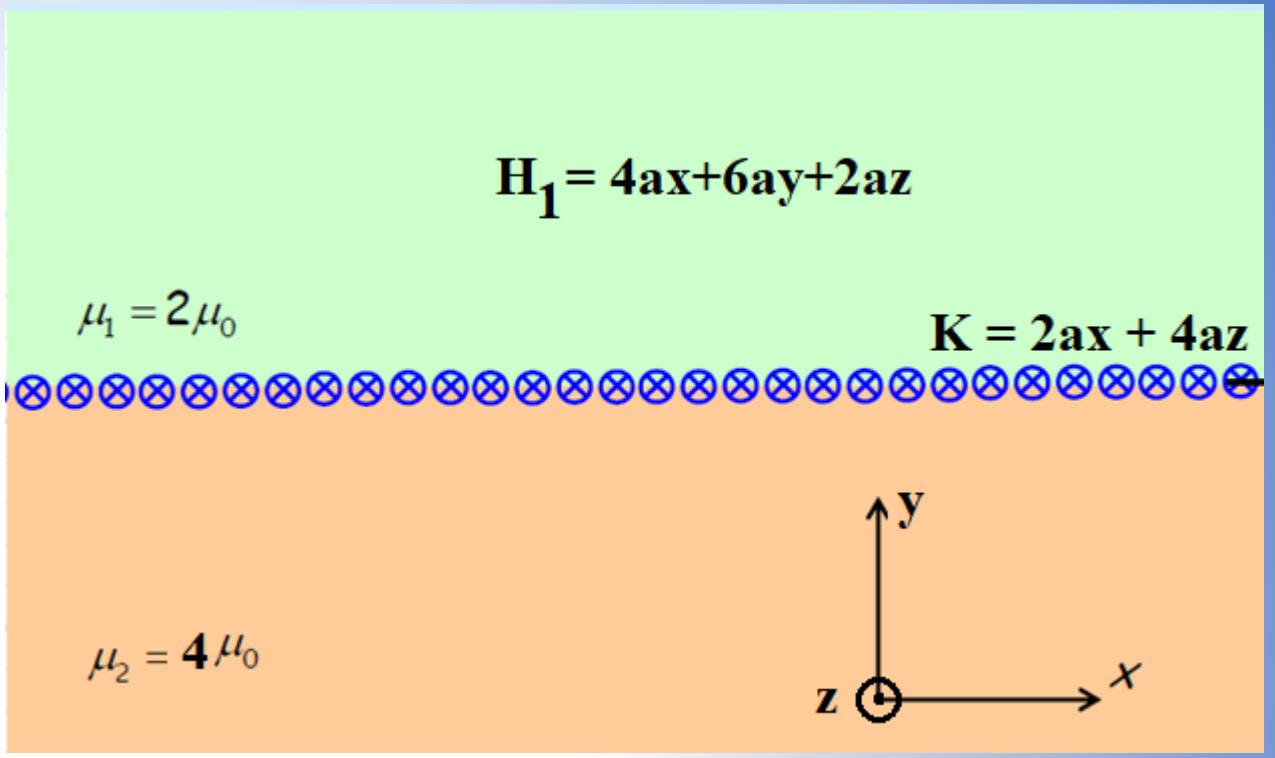
To illustrate these relationships with an example, let us assume that $\mu = \mu_1 = 4\ \mu$H/m in region 1 where $z > 0$, whereas $\mu_2 = 7\ \mu$H/m in region 2 wherever $z < 0$. Moreover, let $\mathbf{K} = 80\mathbf{a}_x$ A/m on the surface $z = 0$. We establish a field, $\mathbf{B}_1 = 2\mathbf{a}_x - 3\mathbf{a}_y + \mathbf{a}_z$ mT, in region 1 and seek the value of $\mathbf{B}_2$.

$$\mathbf{B}_{N2} = \mathbf{B}_{N1} = \mathbf{a}_z \text{ mT}$$

$$\mathbf{H}_{t2} = \mathbf{H}_{t1} - \mathbf{a}_{N12} \times \mathbf{K}$$

$$\mathbf{B}_2 = \mathbf{B}_{N2} + \mathbf{B}_{t2} = 3.5\mathbf{a}_x - 4.69\mathbf{a}_y + \mathbf{a}_z \text{ mT}$$

Problem on boundary condition: Find H_2



y component is Normal component

x & z components are tengential components

For Normal components, $B_{N2} = B_{N1}$ and $H_{N2} = \frac{\mu_1}{\mu_2} H_{N1}$

For Tengential components, $H_{t1} - H_{t2} = K$

$= \frac{2}{4} \cdot 6\hat{a}_y = 3\hat{a}_y$

$\mathbf{H_1 = 4ax+6ay+2az}$ $\quad k = 2\hat{a}_x + 4\hat{a}_2$

For x component

$H_{t_2} = H_{t_1} - a_N \times k$ ← ⊥ to H_{t_1} with k

$= 4\hat{a}_x - (\hat{a}_y \times 4\hat{a}_2) = 0$

For z component

$H_{t_2} = H_{t_1} - a_N \times k$ ← ⊥ to H_{t_1} with k

$= 2\hat{a}_2 - (a_y \times 2\hat{a}_x)$

$= 4\hat{a}_2$

$H_2 = H_{t_2} + H_{N_2} = 3\hat{a}_y + 4\hat{a}_2$


Thank
you